



Apache Airavata Custos

Security & Resource Management Middleware
for Research Computing

Java · Spring Boot · Go · Python · React + TypeScript · Keycloak · Vault



Identity

Persistent researcher
profiles



Authorization

Policy-driven
access control



Allocations

Multi-source compute
credit tracking



Secure Access

Short-lived SSH
certificates

Identity & Authorization



The Identity Challenge

A researcher moves from Georgia Tech to MIT.

New credentials. But their allocations, certificates, and access history should follow them.

Unified Profile: One persistent identity that links all institutional logins. Rich attributes like domain, affiliations, and project memberships feed into policy decisions.



The Authorization Challenge

Dynamic restrictions are the new reality.

Governments and institutions increasingly enforce access rules based on region, affiliation, and clearance, not just roles.

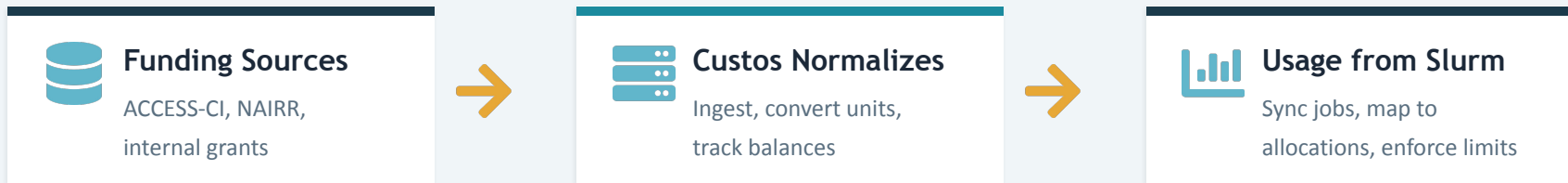
Policy Engine: A general-purpose decision service. Configurable rules, per-tenant policies, audit trails, and dry-run mode.

These two systems work together: the profile provides attributes, the policy engine evaluates them.

Allocation Management

A researcher gets compute time from **ACCESS-CI** (federal), **NAIRR** (AI-focused), and their **university's internal pool**.

Each uses different credit units and protocols. There's no unified view of what's available and what's been consumed.



Key Capabilities

- Credit ledger with immutable charge events
- Award lifecycle management (pending → active → expired)
- PI self-service allocation to CPU, GPU, storage
- Role-based dashboards for PIs, users, and admins

SSH Certificates & Account Provisioning



Short-Lived SSH Certificates

Static SSH keys = key sprawl, no expiration, no audit trail.

Custos replaces them with identity-bound certificates:

1. Researcher authenticates via OIDC
2. Gateway requests certificate from signer
3. Certificate valid for hours, not forever
4. Login node trusts the CA, not individual keys



Account Provisioning

Before signing a certificate, two things must happen:

1. Identity → Account Mapping

CManage Registry maps OIDC identities to Unix usernames. Source of truth for principal validation.

2. Account Creation

Automated provisioning pipeline creates Unix accounts on HPC login nodes via Ansible: home dirs, UIDs, group memberships.

Project Ideas

Students typically take on a focused piece of one project, not the whole thing.

1 Unified Researcher Identity

IDENTITY

Go or Java · REST APIs · Database modeling

Persistent profile across institutions. Identity linking, rich attributes, profile API for downstream services.

2 Dynamic Policy Engine

AUTHORIZATION

Go or Java · Policy engines · Security

Attribute-based access decisions. Per-tenant rules, dry-run mode, audit trail. Integrates OPA or Cedar.

3 Allocation Management

ALLOCATIONS

Go or Java · REST APIs · SQL · Slurm

Credit ledger for multi-source allocations. Ingest from ACCESS-CI/NAIRR, normalize units, enforce limits.

4 Usage Tracking Dashboard

ALLOCATIONS

Go or Java · SQL · React/TypeScript

Sync job data from SlurmDB, map to allocations. Role-based dashboards for PIs, users, and admins.

Project Ideas

continued

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SSH Signing Service

SECURE ACCESS

Go · REST APIs · SSH/Cryptography

Extend the Go signer: certificate search, KRL revocation lists, OpenAPI spec, rate limiting.

6

SSH Signer Admin Portal

SECURE ACCESS

React/TypeScript · Go · REST APIs

Admin UI for client management, certificate browser, CA key rotation, revocation dashboard.

7

Account Provisioning

PROVISIONING

Go or Java · REST APIs · Linux sysadmin

COmanage integration for identity-to-Unix mapping. Ansible-based provisioning pipeline for HPC nodes.

Full project descriptions: github.com/apache/airavata-custos

Get Involved

>_ Explore the Project

Clone the repository

```
git clone git@github.com:apache/airavata-custos.git
```

Start local infrastructure

```
cd compose && docker compose up -d
```

Build and run tests

```
mvn clean install / go build ./...
```

👤 Join the Community

Subscribe to the mailing list

dev@airavata.apache.org

Introduce yourself and ask questions.

Pick an issue and open a PR

Start with a small contribution to get familiar with the codebase and review process.

Real impact

Custos supports national research infrastructure.



github.com/apache/airavata-custos



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